

Alsatian, and Golden Retriever. Consider the class 'Dog' defines a method called `bark()` and a property called `furColor`. Each of its sub-classes (Spitz, Alsatian, and Golden Retriever) will inherit these members. Therefore, the programmer needs to write the code for them just once. Subclasses can also add new members. For instance, take the case of the subclass 'Alsatian'. It can add another method, say, `tremble`. C++ also supports multiple inheritance, where a subclass inherits properties from more than one ancestral class.

1.4 Polymorphism

A Greek term, Polymorphism is the ability to represent oneself in multiple forms. It helps the programmers to treat derived class members, just like their parent class members. By implementing this concept, one can use an operator to perform different operations depending on the operands used.

For instance, if we consider two numbers, the operation 'addition' will generate a sum. Similarly, if the operands are strings, then the operation will produce a third string by concatenation. This phenomenon of making an operator to exhibit various behaviours in different instances is termed as operator overloading.

It is also possible to use same name for different procedures or methods, but the arguments or return types should be unique for each one of them. Different codes are executed accordingly depending on the arguments or the return type.

Let us assume there are three methods of a class sharing the same name 'sum'. One takes two integers as an argument and returns an integer, the other takes three integers as an argument and returns an integer, while the third method takes two floats as an argument and returns a float value. After creating an object in the class, these functions will be called and the